

Date: 07/10Class: VIII

Sarwadhya

$$\begin{aligned}
 & 3xy + 2 + 6y + x \\
 &= 3xy + 6y + 2 + x \\
 &= 3y(x+2) + 1(2+x) \\
 &= (x+2)(3y+1)
 \end{aligned}$$

$$\begin{aligned}
 & 8 - 4a - 2a^3 + a^4 \\
 & 4(2-a) - a^3(2-a)
 \end{aligned}$$

$$1 - 3a - 3az + z$$

$$1 + z - 3a - 3az$$

$$1(1+z) - 3a(1+z)$$

$$(1+z)(1-3a)$$

5. Complete the following activity to find the sum of natural numbers between 1 and 140 which are divisible by 4.

Between 1 and 140, natural numbers divisible by 4

4, 8, ..., 136

How many numbers? $\therefore n = ?$

$n = ?$, $a = 4$, $d = 4$

$$t_n = a + (n-1)d$$

$$136 = 4 + (n-1) \times 4$$

$$n = 34$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{34} = \frac{34}{2} [2 \times 4 + (34-1) \times 4] = \square$$

Sum of numbers from 1 to 140, which are divisible by 4 = $\square$

Date: 07/10

Class: VIII

$$\begin{array}{r}
 1x^2 - 5x + 6 \\
 \swarrow \quad \searrow \\
 x^2 - 3x - 2x + 6 \\
 x(x-3) - 2(x-3) \\
 (x-3)(x-2)
 \end{array}$$

$$\begin{array}{r}
 1a^2 - 1a - 6 \\
 a^2 - 3a + 2a - 6 \\
 a(a-3) + 2(a-3) \\
 (a-3)(a+2)
 \end{array}$$

$$\begin{array}{r}
 2x^2 + x - 3 \\
 2x^2 + 3x - 2x - 3 + 3 \\
 x(2x+3) - 1(2x+3) \\
 (2x+3)(x-1)
 \end{array}$$

Date: 07/10Sum of first 55 terms in an A.P. is 3300, find its 28th term.

Class: _____

$$S_{55} = 3300$$

$$n = 55$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{55} = \frac{55}{2} [2a + 54d]$$

$$\begin{array}{r} 60 \\ 3300 \times 2 \\ \hline 55 \end{array} = 2a + 54d$$

$$11, 5$$

$$60 \times 2 = 2(a + 27d)$$

$$a + 27d = \frac{60 \times 2}{2}$$

$$a + 27d = 60$$

Date: 07/10

$$(6) \frac{1}{3}x + y = \frac{10}{3}; 2x + \frac{1}{4}y = \frac{11}{4}$$

Class: _____

$$\frac{1}{3}x + \frac{y \times 3}{1 \times 3} = \frac{10}{3}$$

$$\frac{1}{\cancel{3}}x + \frac{\cancel{3}}{\cancel{3}}y = \frac{\cancel{10}}{\cancel{3}}$$

$$x + 3y = 10$$

$$\frac{4 \times 2}{4 \times 1}x + \frac{1}{4}y = \frac{11}{4}$$

$$\frac{\cancel{8}}{\cancel{4}}x + \frac{\cancel{1}}{\cancel{4}}y = \frac{\cancel{11}}{\cancel{4}}$$

$$8x + y = 11$$

$$x = 3$$

$$y = 1$$